24/01/2025
Evening

Aakash
Medical | IIT-JEE | Foundations

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Answers & Solutions

Time : 3 hrs. *for* M.M. : 300

JEE (Main)-2025 Phase-1

[Computer Based Test (CBT) mode]

(Mathematics, Physics and Chemistry)

IMPORTANT INSTRUCTIONS:

(1) The test is of **3 hours** duration.

(2) This test paper consists of 75 questions. Each subject (MPC) has 25 questions. The maximum marks are 300.

(3) This question paper contains **Three** Parts. **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is **Mathematics**. Each part has only two sections: **Section-A** and **Section-B**.

(4) **Section - A :** Attempt all questions.

(5) **Section - B :** Attempt all questions.

(6) **Section - A (01 – 20)** contains 20 multiple choice questions which have **only one correct answer**. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

(7) **Section - B (21 – 25) contains** 5 **Numerical value** based questions. The answer to each question should be rounded off to the **nearest integer**. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.



MATHEMATICS

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer:

1. Group A consists of 7 boys and 3 girls, while group B consists of 6 boys and 5 girls. The number of ways, 4 boys and 4 girls can be invited for a picnic if 5 of them must be from group A and the remaining 3 from group B, is equal to:

(1) 8575 (2) 8750

(3) 8925 (4) 9100

Answer (3)

Sol.

Group A		Group B		
B	G	B	G	Ways
4	1	0	3	${}^7C_4 \cdot {}^3C_1 \cdot {}^6C_0 \cdot {}^5C_3$
3	2	1	2	${}^7C_3 \cdot {}^3C_2 \cdot {}^6C_1 \cdot {}^5C_2$
2	3	2	1	${}^7C_2 \cdot {}^3C_3 \cdot {}^6C_2 \cdot {}^5C_1$

Total ways = $30 \cdot {}^7C_4 + 180 \cdot {}^7C_3 + 75 \cdot {}^7C_2 = 8925$

2. Let $\vec{a} = 3\hat{i} - \hat{j} + 2\hat{k}$, $\vec{b} = \vec{a} \times (\hat{i} - 2\hat{k})$ and $\vec{c} = \vec{b} \times \hat{k}$.

Then the projection of $\vec{c} = 2\hat{j}$ on $\vec{a}$ is:

(1) $\sqrt{14}$ (2) $2\sqrt{7}$

(3) $2\sqrt{14}$ (4) $3\sqrt{7}$

Answer (3)

Sol.

$$\vec{a} = 3\hat{i} - \hat{j} + 2\hat{k} \quad \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -1 & 2 \\ 1 & 0 & -2 \end{vmatrix} = 2\hat{i} + 8\hat{j} + \hat{k}$$

$$\vec{b} = \vec{a} \times (\hat{i} - 2\hat{k}) \Rightarrow$$

$$\vec{c} = \vec{b} \times \hat{k} = 8\hat{i} - 2\hat{j} \Rightarrow \vec{c} - 2\hat{j} = 8\hat{i} - 4\hat{j}$$

Projection of $(\vec{c} - 2\hat{j})$ on $\vec{a}$

$$= (\vec{c} - 2\hat{j}) \cdot \frac{\vec{a}}{|\vec{a}|} = \left(\frac{24+4}{\sqrt{14}}\right) = 2\sqrt{14}$$

3. Let $f: (0, \infty) \to \mathbf{R}$ be function which is differentiable at all points of its domain and satisfies the condition $x^2 f'(x) = 2xf(x) + 3$, with $f(1) = 4$. Then $2f(2)$ is equal to:

(1) 23 (2) 19

(3) 29 (4) 39

Answer (4)

Sol. $x^2 f'(x) = 2xf(x) + 3$

Let $y = f(x) \Rightarrow \frac{dy}{dx} = f'(x)$

$$\frac{dy}{dx} = \frac{2y}{x} + \frac{3}{x^2}$$

$$\Rightarrow \frac{dy}{dx} + y\left(\frac{-2}{x}\right) = \frac{3}{x^2}$$

$$\text{I.F} = e^{\int \frac{-2}{x} dx} = e^{-2\ln x} = \frac{1}{x^2}$$

$$\Rightarrow y\left(\frac{1}{x^2}\right) = \int \frac{3}{x^2} \times \frac{1}{x^2} dx + c$$

$$\frac{y}{x^2} = -x^{-3} + c, \; f(1) = 4 = y$$

$$\frac{4}{(1)^2} = -(1)^{-3} + c \Rightarrow c = 5$$

$$\Rightarrow \frac{y}{x^2} = \frac{-1}{x^3} + 5 \Rightarrow f(x) = \frac{-1}{x} + 5x^2$$

$$2f(2) = 2\left[\frac{-1}{2} + 5(2)^2\right] = 2\left[\frac{-1}{2} + 20\right] = 39$$



4. Let the position vectors of three vertices of a triangle be $4\vec{p}+\vec{q}-3\vec{r}$, $-5\vec{p}+\vec{q}+2\vec{r}$ and $2\vec{p}-\vec{q}+2\vec{r}$. If the position vectors of the orthocentre and the circumcentre of the triangle are $\frac{\vec{p}+\vec{q}+\vec{r}}{4}$ and $\alpha\vec{p}+\beta\vec{q}+\gamma\vec{r}$ respectively, then $\alpha + 2\beta + 5\gamma$ is equal to:

(1) 4 (2) 6

(3) 3 (4) 1

Answer (3)

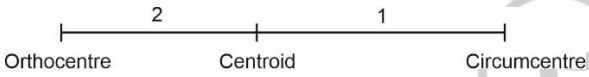
Sol. Orthocentre $=\frac{\vec{p}+\vec{q}+\vec{r}}{4}$

Circumcentre $=\alpha\vec{p}+\beta\vec{q}+\gamma\vec{r}$

Centroid $=\left(\frac{4\vec{p}+\vec{i}-3\vec{r}-5\vec{p}+\vec{q}+2\vec{r}+2\vec{p}-\vec{q}+2\vec{r}}{3}\right)$

$=\left(\frac{\vec{p}+\vec{q}+\vec{r}}{3}\right)$

Now



$$\frac{2(\alpha\vec{p}+\beta\vec{q}+\gamma\vec{r})+\frac{\vec{p}+\vec{q}+\vec{r}}{4}}{2+1}=\left(\frac{\vec{p}+\vec{q}+\vec{r}}{3}\right)$$

$\Rightarrow 8(\alpha\vec{p}+\beta\vec{q}+\gamma\vec{r})=3(\vec{p}+\vec{q}+\vec{r})$

$\Rightarrow 8\alpha=3,\ 8\beta=3,\ 8\gamma=3$

$\alpha=\frac{3}{8}=\beta=\gamma$

$\alpha+2\beta+5\gamma=\frac{3}{8}+\frac{6}{8}+\frac{15}{8}=\frac{27}{8}=3$

5. Let (2, 3) be the largest open interval in which the function $f(x) = 2\log_e(x-2) - x^2 + ax + 1$ is strictly increasing and (b, c) be the largest open interval, in which the function $g(x) = (x-1)^3(x+2-a)^2$ is strictly decreasing. Then $100(a+b-c)$ is equal to:

(1) 280 (2) 360

(3) 160 (4) 420

Answer (2)

Sol. $f(x) = 2\ln(x-2) - x^2 + ax + 1$

$$f'(x)=\frac{2}{x-2}-2x+a>0$$

$$f''(x)=\frac{-2}{(x-2)^2}-2<0$$

$\Rightarrow$ $f'(x)$ is decreasing

and $f'(3)\geq 0$

$2-6+a\geq 0 \Rightarrow a\geq 4$

$a_{min} = 4$

$\because$ $g(x)$ is strictly decreasing

Now, $g(x) = (x-1)^3(x-2)^2$

$g'(x)=2(x-1)^3(x-2)+3(x-1)^2(x-2)^2<0$

$=(x-1)^2(x-2)[2(x-1)+3(x-2)]<0$

$=(x-1)^2(x-2)(5x-8)<0$

$\Rightarrow x\in\left(\frac{8}{5},2\right) \Rightarrow b=\frac{8}{5}\quad c=2$

Now, $100(a+b-c)=100\left[4+\frac{8}{5}-2\right]$

$=100\times\frac{18}{5}=360$

6. The equation of the chord, of the ellipse $\frac{x^2}{25}+\frac{y^2}{16}=1$, whose mid-point is (3, 1) is:

(1) $5x + 16y = 31$

(2) $25x + 101y = 176$

(3) $48x + 25y = 169$

(4) $4x + 122y = 134$

Answer (3)

Sol. $\frac{x^2}{25}+\frac{y^2}{16}=1$

Chord with mid-point (3, 1)

$T = S_1$



$$\frac{xx_1}{25}+\frac{yy_1}{16}-1=\frac{x_1^2}{25}+\frac{y_1^2}{16}$$

$$\frac{3x}{25}+\frac{y}{16}=\frac{9}{25}+\frac{1}{16}$$

$$48x + 25y = 9\times16 + 25$$

$$48x + 25y = 169$$

7. Let $[x]$ denote the greatest integer function, and let m and n respectively be the numbers of the points, where the function $f(x) = [x] + |x-2|$, $-2 < x < 3$, is not continuous and not differentiable. Then $m + n$ is equal to :

(1) 8

(2) 6

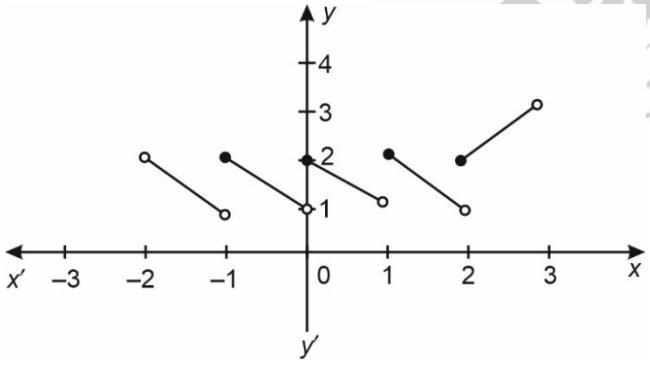
(3) 7

(4) 9

Answer (1)

Sol. $f(x) = [x] + |x-2|$, $-2 < x < 3$

$$\therefore\ f(x)=\begin{cases}-x, & -2<x<-1\\ 1-x, & -1\le x<0\\ 2-x, & 0\le x<1\\ 3-x, & 1\le x<2\\ x, & 2\le x<3\end{cases}$$



It is clearly discontinues at 4 points and non-differentiable at 4 points.

$\therefore\ m + n = 8$

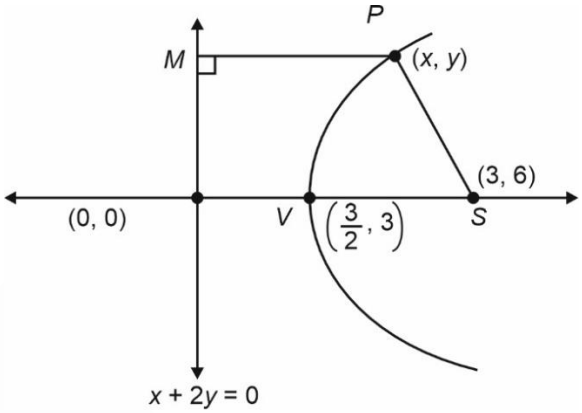
8. If the equation of the parabola with vertex $V\left(\frac{3}{2}, 3\right)$ and the directrix $x + 2y = 0$ is $\alpha x^2 + \beta y^2 - \gamma xy - 30x - 60y + 225 = 0$, then $\alpha + \beta + \gamma$ is equal to:

(1) 7 (2) 6

(3) 9 (4) 8

Answer (3)

Sol.



$\therefore$ Equation of parabola is $PS = (1)\ PM$

$\Rightarrow PS^2 = PM^2$

$$(x-3)^2+(y-6)^2=\left(\frac{x+2y}{\sqrt{5}}\right)^2$$

$$5x^2-30x+5y^2-60y+225=x^2+4y^2+4xy$$

$$4x^2+y^2-4xy-30x-60y+225=0$$

We get : $\alpha = 4$, $\beta = 1$, $\gamma = 4$

$\therefore\ \alpha + \beta + \gamma = 9$

9. Let $A = [a_{ij}]$ be a square matrix of order 2 with entries either 0 or 1. Let E be the event that A is an invertible matrix. Then the probability $P(E)$ is :

(1) $\frac{3}{16}$ (2) $\frac{3}{8}$

(3) $\frac{1}{8}$ (4) $\frac{5}{8}$

Answer (2)

Sol. $A = [a_{ij}]_{2\times2}$ and entries are 0 or 1.



$\therefore \begin{vmatrix} a & b \\ c & d \end{vmatrix} = 0$

$\Rightarrow ad - bc = 0$

Case I: $ad = bc = 1$

$\therefore a = b = c = d = 1$

Case II: $ad = bc = 0$

$a = 0, d = 0$ $\quad b = 0, c = 0$

$a = 0, d = 1$ $\quad b = 0, c = 1$

$a = 1, d = 0$ $\quad b = 1, c = 0$

$\therefore$ Total 10 cases when matrix is non invertible

Total possible matrix = $2^4 = 16$

Required probability of invertible

$= \frac{16-10}{16} = \frac{6}{16} = \frac{3}{8}$

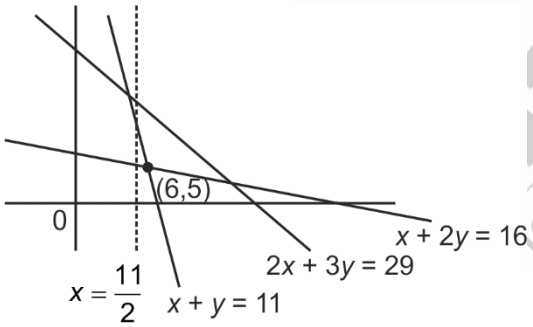
10. Let the points $\left(\frac{11}{2}, \alpha\right)$ lie on or inside the triangle with sides $x + y = 11$, $x + 2y = 16$ and $2x + 3y = 29$. Then the product of the smallest and the largest value of α is equal to :

(1) 22 (2) 55

(3) 33 (4) 44

Answer (3)

Sol.



Clearly, $x = \frac{11}{2}$ intersect $x + y - 11 = 0$ at $\left(\frac{11}{2}, \frac{11}{2}\right)$

and $2x + 3y - 29 = 0$ at $\left(\frac{11}{2}, 6\right) \Rightarrow \alpha = \left[\frac{11}{2}, 6\right]$

$\alpha_{min} \cdot \alpha_{max} = \frac{11}{2} \cdot 6 = 33$

11. If $7 = 5 + \frac{1}{7}(5 + \alpha) + \frac{1}{7^2}(5 + 2\alpha) + \frac{1}{7^3}(5 + 3\alpha) +$ ... ∞, then the value of α is

(1) 1

(2) 6

(3) $\frac{1}{7}$

(4) $\frac{6}{7}$

Answer (2)

Sol. $S = a + (a + d)r + (a + 2d)r^2 + \ldots$

Then $S = \frac{a}{1-r} + \frac{dr}{(1-r)^2}, |r| < 1$

Since, $r = \frac{1}{7}$ and $a = 5$, $d = \alpha$

$7 = \frac{5}{1 - \frac{1}{7}} + \frac{\alpha \cdot \frac{1}{7}}{\left(1 - \frac{1}{7}\right)^2}$

$\Rightarrow \alpha = 6$

12. Suppose A and B are the coefficients of 30th and 12th terms respectively in the binomial expansion of $(1 + x)^{2n-1}$. If $2A = 5B$, then n is equal to :

(1) 19 (2) 20

(3) 21 (4) 22

Answer (3)

Sol. $T_{r+1} = {}^{2n-1}C_r \, x^r$

Coefficient of $T_{30} = {}^{2n-1}C_{29} = A$

Coefficient of $T_{12} = {}^{2n-1}C_{11} = B$

$\Rightarrow 2\left({}^{2n-1}C_{29}\right) = 5\left({}^{2n-1}C_{11}\right)$

$\frac{2n}{30} \cdot {}^{2n-1}C_{29} = \frac{2n}{12} \cdot {}^{2n-1}C_{11}$

$\Rightarrow n = 21$



13. Let

$$A=\left\{x\in(0,\pi)-\left(\frac{\pi}{2}\right):\log_{(2/\pi)}|\sin x|+\log_{(2/\pi)}|\cos x|=2\right\}$$

and $B=\{x\geq 0:\sqrt{x}\,(\sqrt{x}-4)-3|\sqrt{x}-2|+6=0\}$.

Then $n(A\cup B)$ is equal to

(1) 2 (2) 6

(3) 4 (4) 8

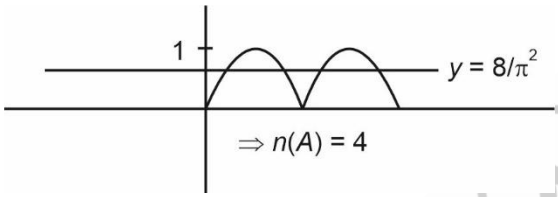
Answer (4)

Sol. $A=\left\{x\in(0,\pi)-\left(\frac{\pi}{2}\right):\log_{(2/\pi)}|\sin x|+\log_{(2/\pi)}|\cos x|=2\right\}$

$\log_{2/x}|\sin x\cos x|=2$

$\frac{|\sin 2x|}{2}=\frac{4}{\pi^2}$

$\Rightarrow |\sin 2x|=\frac{8}{\pi^2}$



and $B=\left\{x\geq 0:\sqrt{x}(\sqrt{x}-4)-3|\sqrt{x}-2|+6=0\right\}$

$x-4\sqrt{x}-3\sqrt{x}+6+6=0$ when $\sqrt{x}-2\geq 0$

$\sqrt{x}\geq 2$

$\Rightarrow x-7\sqrt{x}+12=0$

$\Rightarrow x-4\sqrt{x}-3\sqrt{x}+12=0$

$\Rightarrow \sqrt{x}\sqrt{x}-4)-3(\sqrt{x}-4)=0$

$\sqrt{x}=4,\sqrt{x}=3$

$\Rightarrow x=16,9$

When $\sqrt{x}-2<0$

$\Rightarrow x-4\sqrt{x}+3\sqrt{x}-6+6=0$

$\Rightarrow x-\sqrt{x}=0$

$\Rightarrow \sqrt{x}(\sqrt{x}-1)=0$

$\Rightarrow \sqrt{x}=0,1$

$\Rightarrow x=0,1$

$\Rightarrow n(B)=4$

Now as A and B are mutually exclusive sets

$N(A\cup B)=4+4=8$

14. In an arithmetic progression, if $S_{40}=1030$ and $S_{12}=57$, then $S_{30}-S_{10}$ is equal to

(1) 525 (2) 510

(3) 515 (4) 505

Answer (3)

Sol. $S_{40}=1030 \Rightarrow \frac{40}{2}[2a+39d]=1030$

$\Rightarrow 2a+39d=\frac{103}{2}$ ...(1)

$S_{12}=57 \Rightarrow \frac{12}{2}[2a+11d]=57$

$\Rightarrow 2a+11d=\frac{57}{6}$ ...(2)

Equation (1) – equation (2)

$28d=\frac{103}{2}-\frac{57}{6}$

$28d=\frac{309-57}{6}$

$d=\frac{3}{2}$

$\Rightarrow a=-\frac{7}{2}$

$S_{30}-S_{10}=\frac{30}{2}[2a+29d]-\frac{10}{2}[2a+9d]$

$=15[2a+29d]-5[2a+9d]$

$=5[6a+87d-2a-9d]$

$=5[4a+78d]$

$=5[-14+117]$

$=515$



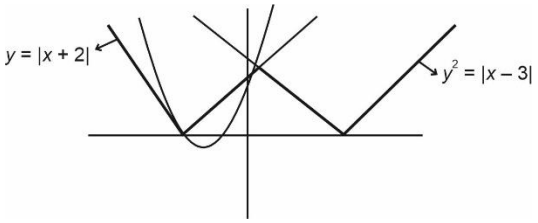
15. The number of real solution(s) of the equation $x^2 + 3x + 2 = \min\{|x-3|, |x+2|\}$ is

(1) 2 (2) 1

(3) 3 (4) 0

Answer (1)

Sol. $x^2 + 3x + 2 = \min\{|x-3|, |x+2|\}$



$y = x^2 + 3x + 2$

$$y = x^2 + 2\left(\frac{3}{2}\right)x + \frac{9}{4} - \frac{9}{4} + 2$$

$$y = \left(x + \frac{3}{2}\right)^2 - \frac{1}{4}$$

$$y + \frac{1}{4} = \left(x + \frac{3}{2}\right)^2$$

$\Rightarrow$ Parabola vertex $\left(\frac{-3}{2}, \frac{-1}{4}\right)$

$\Rightarrow$ By graph 2 solution possible

16. For some a, b, let $f(x) = \begin{vmatrix} a + \frac{\sin x}{x} & 1 & b \\ a & 1 + \frac{\sin x}{x} & b \\ a & 1 & b + \frac{\sin x}{x} \end{vmatrix}$, $x \neq 0$, $\lim\limits_{x \to 0} f(x) = \lambda + \mu a + \nu b$. Then $(\lambda + \mu + \nu)^2$ is equal to

(1) 36

(2) 16

(3) 9

(4) 25

Answer (2)

Sol. $\lim\limits_{x \to 0} \begin{vmatrix} a + \frac{\sin x}{x} & 1 & b \\ a & 1 + \frac{\sin x}{x} & b \\ a & 1 & b + \frac{\sin x}{x} \end{vmatrix} = \lambda + \mu a + \nu b$

At lim $x \to 0$,

$$f(x) = \begin{vmatrix} a+1 & 1 & b \\ a & 1+1 & b \\ a & 1 & b+1 \end{vmatrix} = \lambda + \mu a + \nu b$$

$R_1 \to R_1 - R_2$

$R_2 \to R_2 - R_3$

$$\begin{vmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ a & 1 & b+1 \end{vmatrix} = \lambda + \mu a + \nu b$$

$C_2 \to C_1 - C_2$

$$\begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ a & a+1 & b+1 \end{vmatrix} = \lambda + \mu a + \nu b$$

$a + b + 2 = \lambda + \mu a + \nu b$

$\lambda = 2, \ \mu = 1, \ \nu = 1$

$(\lambda + \mu + \nu) = (2 + 1 + 1)^2 = 16$

17. The function $f : (-\infty, \infty) \to (-\infty, 1)$, defined by $f(x) = \frac{2^x - 2^{-x}}{2^x + 2^{-x}}$ is:

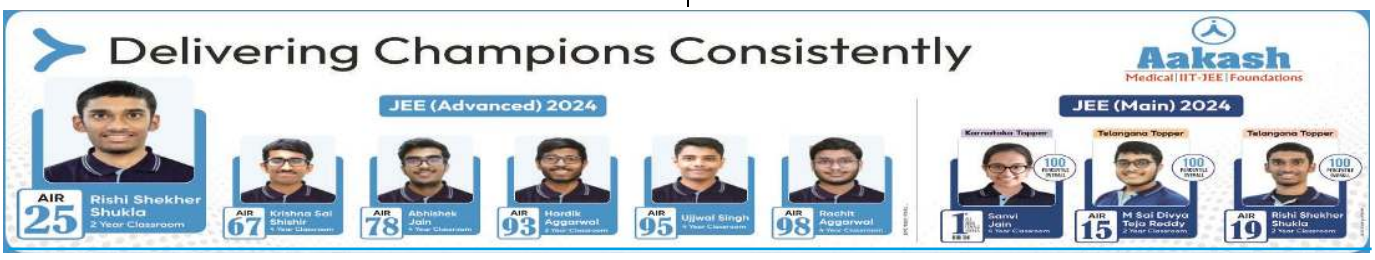
(1) Onto but not one-one

(2) Both one-one and onto

(3) One-one but not onto

(4) Neither one-one nor onto

Answer (3)



Sol. $f(x)=\dfrac{2^x-2^{-x}}{2^x+2^{-x}}=\dfrac{2^{2x}-1}{2^{2x}+1}=\dfrac{2^{2x}+1-2}{2^{2x}+1}$

$$f(x)=1-\frac{2}{2^{2x}+1}$$

2^{2x} is one-one, so $f(x)$ is one-one

for $x \in (-\infty, \infty)$

$2^x \in (0, \infty)$

$2^{2x}+1 \in (1, \infty)$

$$\frac{1}{2^{2x}+1} \in (1, 0)$$

$$\frac{-2}{2^{2x}+1} \in (-2, 0)$$

$$1-\frac{2}{2^{2x}+1} \in (-1, 1)$$

Range of $f(x)$ is $(-1, 1)$ but codomain of $f(x)$ is $(-\infty, 1)$

So $f(x)$ is into

18. The area of the region enclosed by the curves $y = e^x$, $y = |e^x - 1|$ and y-axis is:

(1) $1 + \log_e 2$

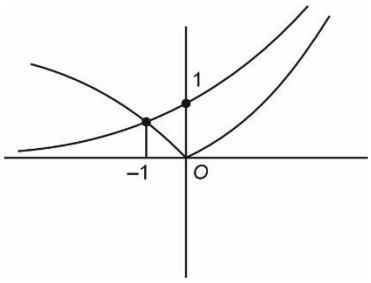
(2) $\log_e 2$

(3) $1 - \log_e 2$

(4) $2\log_e 2 - 1$

Answer (3)

Sol.



$e^x = 1 - e^x \Rightarrow 2e^x = 1$

$$\Rightarrow e^x = \frac{1}{2}$$

$$\Rightarrow x = \ln\frac{1}{2}$$

$$\int_{\ln(1/2)}^{0}\left[e^x-(1-e^x)\right]dx$$

$$=\int_{\ln 2}^{0}(2e^x-1)dx=2e^x-x\Big|_{-\ln 2}^{0}$$

$= 2 - (1 + \ln 2)$

$= 1 - \log_e 2$

19. If $\alpha > \beta > \gamma > 0$, then the expression

$$\cot^{-1}\left\{\beta+\frac{(1+\beta^2)}{(\alpha-\beta)}\right\}+\cot^{-1}\left\{\gamma+\frac{(1+\gamma^2)}{(\beta-\gamma)}\right\}+\cot^{-1}\left\{\alpha+\frac{1+\alpha^2}{\gamma-\alpha}\right\}$$ is equal to:

(1) 0

(2) $\dfrac{\pi}{2}-(\alpha+\beta+\gamma)$

(3) 3π

(4) π

Answer (4)

Sol. $$\cot^{-1}\left(\frac{\alpha\beta-\beta^2+1+\beta^2}{\alpha-\beta}\right)+\cot^{-1}\left(\frac{\beta\gamma-\gamma^2+1+\gamma^2}{\beta-\gamma}\right)+\cot^{-1}\left(\frac{\alpha\gamma-\alpha^2+1+\alpha^2}{\gamma-\alpha}\right)$$

$$=\cot^{-1}\left(\frac{1+\alpha\beta}{\alpha-\beta}\right)+\cot^{-1}\left(\frac{1+\beta\gamma}{\beta-\gamma}\right)+\cot^{-1}\frac{1+\gamma\alpha}{\gamma-\alpha}$$

$$=\tan^{-1}\left(\frac{\alpha-\beta}{1+\alpha\beta}\right)+\tan^{-1}\left(\frac{\beta-\gamma}{1+\beta\gamma}\right)+\pi+\tan^{-1}\left(\frac{\gamma-\alpha}{1+\gamma\alpha}\right)$$



$= \tan^{-1}\alpha - \tan^{-1}\beta + \tan^{-1}\beta - \tan^{-1}\gamma + \pi + \tan^{-1}\gamma - \tan^{-1}\alpha$

$= \pi$

20. If the system of equations

$x + 2y - 3z = 2$

$2x + \lambda y + 5z = 5$

$14x + 3y + \mu z = 33$

has infinitely many solutions, then $\lambda + \mu$ is equal to:

(1) 12

(2) 10

(3) 13

(4) 11

Answer (1)

Sol. $\Delta = \begin{vmatrix} 1 & 2 & -3 \\ 2 & \lambda & 5 \\ 14 & 3 & \mu \end{vmatrix} = 0 \Rightarrow \lambda\mu + 42\lambda - 4\mu + 107 = 0$

$\Delta_1 = \begin{vmatrix} 2 & 2 & -3 \\ 5 & \lambda & 5 \\ 33 & 3 & \mu \end{vmatrix} = 0 \Rightarrow 2\lambda\mu + 99\lambda - 10\mu + 255 = 0$

$\Delta_2 = \begin{vmatrix} 1 & 2 & -3 \\ 2 & 5 & 5 \\ 14 & 33 & \mu \end{vmatrix} = 0 \Rightarrow \mu = 13$

Also, $\lambda = -1$

Hence, $\lambda + \mu = 13 - 1 = 12$

SECTION - B

Numerical Value Type Questions: This section contains 5 Numerical based questions. The answer to each question should be rounded-off to the nearest integer.

21. Let P be the image of the point $Q(7, -2, 5)$ in the line $L : \frac{x-1}{2} = \frac{y+1}{3} = \frac{z}{4}$ and $R(5, p, q)$ be a point on L.

Then the square of the area of ΔPQR is _____.

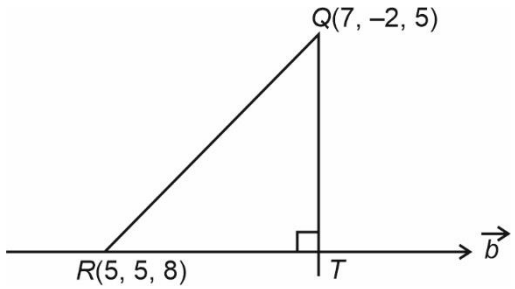
Answer (957)

Sol. Let $R(2\lambda + 1, 3\lambda - 1, 4\lambda)$

$2\lambda + 1 = 5$

$\lambda = 2$

$\Rightarrow \quad R(5, 5, 8)$



Let say $T(2\lambda + 1, 3\lambda - 1, 4\lambda)$

So, $\overrightarrow{QT} = (2\lambda - 6)\hat{i} + (3\lambda + 1)\hat{j} + (4\lambda - 5)\hat{k}$

and $\vec{b} = 2\hat{i} + 3\hat{j} + 4\hat{k}$

$\overrightarrow{QT} \cdot \vec{b} = 0$

$4\lambda - 12 + 9\lambda + 3 + 16\lambda - 20 = 0$

$\lambda = 1$

$T(3, 2, 4)$

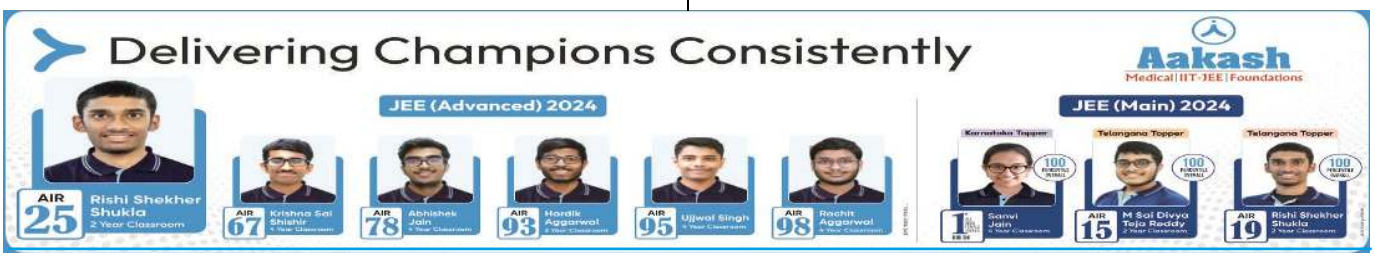
$QT = \sqrt{33}, \; RT = \sqrt{29}$

$(\text{ar}(\Delta PQR))^2 = \left(\frac{1}{2}\sqrt{33} \cdot 2 \cdot \sqrt{29}\right)^2$

$= 957$

22. If $\int \frac{2x^2 + 5x + 9}{\sqrt{x^2 + x + 1}} dx = x\sqrt{x^2 + x + 1} + \alpha\sqrt{x^2 + x + 1} + \beta \log_e \left| x + \frac{1}{2} + \sqrt{x^2 + x + 1} \right| + C$, where C is the constant of integration, then $\alpha + 2\beta$ is equal to _____.

Answer (16)



Sol. $I=\int\frac{2x^2+5x+9}{x^2+x+1}dx$

Let $\frac{2x^2+5x+9}{\sqrt{x^2+x+1}}=\frac{A\left(x^2+x+1\right)+B(2x+1)+C}{\sqrt{x^2+x+1}}$

Then, $A=2, B=\frac{3}{2}$ and $C=\frac{11}{2}$

$\therefore\ I=\int\frac{2\left(x^2+x+1\right)+\frac{3}{2}(2x+1)+\frac{11}{2}}{\sqrt{x^2+x+1}}dx$

$=2\int\sqrt{x^2+x+1}\,dx+\frac{3}{2}\cdot2\sqrt{x^2+x+1}+\frac{11}{2}\int\frac{1}{\sqrt{x^2+x+1}}dx$

$=2\int\sqrt{\left(x+\frac{1}{2}\right)^2+\left(\frac{\sqrt{3}}{2}\right)^2}dx+3\sqrt{x^2+x+1}+\frac{11}{2}\int\frac{dx}{\left(x+\frac{1}{2}\right)^2+\left(\frac{\sqrt{3}}{2}\right)^2}$

$\left\{\frac{x+\frac{1}{2}}{2}\sqrt{x^2+x+1}+\frac{3}{8}\ln\left|\left(x+\frac{1}{2}\right)+\sqrt{x^2+x+1}\right|\right\}+3\sqrt{x^2+x+1}+\frac{11}{2}\ln\left|x+\frac{1}{2}+\sqrt{x^2+x+1}\right|+C$

$=\left(\frac{2x+1}{2}\right)\sqrt{x^2+x+1}+\frac{3}{4}\ln\left|x+\frac{1}{2}+\sqrt{x^2+x+1}\right|+3\sqrt{x^2+x+1}+\frac{22}{7}\left|x+\frac{1}{2}+\sqrt{x^2+x+1}\right|+C$

$=\frac{2x+7}{2}\sqrt{x^2+x+1}+\frac{25}{4}\ln\left|x+\frac{1}{2}+\sqrt{x^2+x+1}\right|+C$

$=x\sqrt{x^2+x+1}+\frac{7}{2}\sqrt{x^2+x+1}+\frac{25}{4}\ln\left|x+\frac{1}{2}+\sqrt{x^2+x+1}\right|+C$

$\therefore\ \alpha=\frac{7}{2},\beta=\frac{25}{4}$

Then $\alpha+2\beta=\frac{7}{2}+\frac{25}{4}=16$

23. Let $H_1:\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ and $H_2:-\frac{x^2}{A^2}+\frac{y^2}{B^2}=1$ be two hyperbolas having length of latus rectums $15\sqrt{2}$ and $12\sqrt{5}$ respectively. Let their eccentricities be $e_1=\sqrt{\frac{5}{2}}$ and e_2 respectively. If the product of the lengths of their transverse axes is $100\sqrt{10}$, then $25e_2^2$ is equal to _____.

Answer (55)

Sol. $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$

$\frac{2b^2}{a}=15\sqrt{2}$ ...(i)

$\sqrt{1+\frac{b^2}{a^2}}=\sqrt{\frac{5}{2}}$ ...(ii)

From (i) and (ii)

$a=5\sqrt{2}$ and $b^2=75$

$\frac{x^2}{A^2}-\frac{y^2}{B^2}=-1$

$\frac{2A^2}{B}=12\sqrt{5}$ ...(iii)



Since, product of transverse axis is = $100\sqrt{10}$

$(2A)\cdot(2B) = 100\sqrt{10}$ ...(iv)

From (iii) and (iv)

$A^2 = 150$ and $B = 5\sqrt{5}$

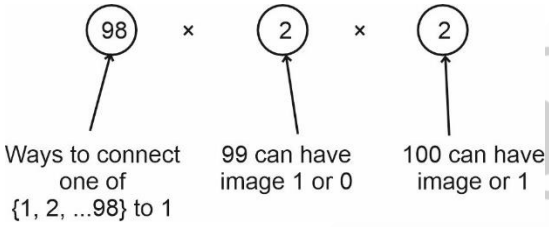
$$e_2 = \sqrt{1+\frac{A^2}{B^2}} = \sqrt{\frac{11}{5}}$$

$$\therefore \quad 25e_2^2 = 25\left(\frac{11}{5}\right) = 55$$

24. Number of functions $f : \{1, 2, ..., 100\} \to \{0, 1\}$, that assign 1 to exactly one of the positive integers less than or equal to 98, is equal to _____.

Answer (392)

Sol.



$= 98 \times 2 \times 2 = 392$

25. Let $y = y(x)$ be the solution of the differential equation $2\cos x\frac{dy}{dx} = \sin 2x - 4y\sin x,\ x \in \left(0, \frac{\pi}{2}\right)$. If $y\left(\frac{\pi}{3}\right) = 0$, then $y'\left(\frac{\pi}{4}\right) + y\left(\frac{\pi}{4}\right)$ is equal to_____.

Answer (01)

Sol. $2\cos x\frac{dy}{dx} = \sin 2x - 4y\sin x$

$$\frac{dy}{dx} = \frac{2\sin x\cos x}{2\cos x} - \frac{4y\sin x}{2\cos x}$$

$$\frac{dy}{dx} + 2y\tan x = \sin x$$

I.F $= e^{\int 2\tan x dx}$

I.F $= e^{2\ln|\sec x|}$

I.F $= |\sec x|^2 = \sec^2 x\ \forall\ x \in (0, \pi/2)$

$$y.\sec^2 x = \int \sin x.\sec^2 x dx$$

$$y.\sec^2 x = \int \frac{\sin x}{\cos^2 x}dx$$

Let, $\cos x = t$

$$-\sin x dx = dt$$

$$y.\sec^2 x = -\int \frac{1}{t^2}dt$$

$$y.\sec^2 x = \frac{1}{t} + C$$

$y.\sec^2 x = \sec x + C$

$y(\pi/3) = 0$

$C = -2$

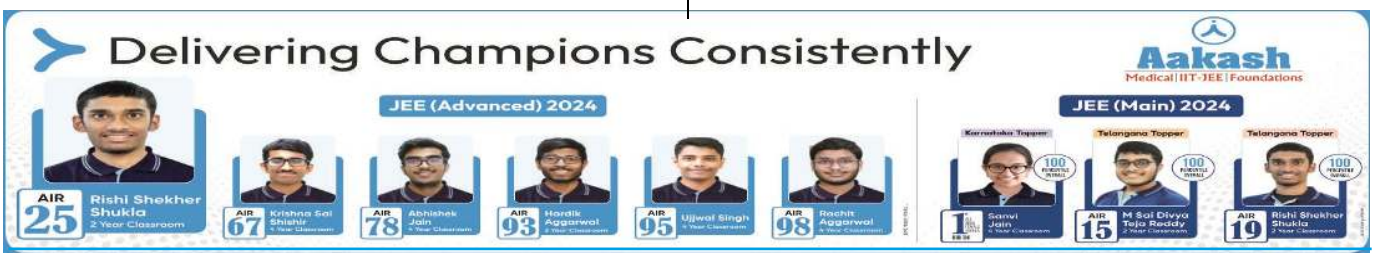
$\therefore\ y = \cos x - 2\cos^2 x$

$y'(x) = -\sin x + 2\sin 2x$

$$y'\left(\pi/4\right) = \frac{1}{\sqrt{2}} + 2$$

$$y'(\pi/4) = \frac{1}{\sqrt{2}} - 1$$

$$\therefore\ y'\left(\frac{\pi}{4}\right) + y\left(\frac{\pi}{4}\right) = 1$$



PHYSICS

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer :

26. The temperature of a body in air falls from 40°C to 24°C in 4 minutes. The temperature of the air is 16°C. The temperature of the body in the next 4 minutes will be :

(1) $\frac{14}{3}$°C (2) $\frac{28}{3}$°C

(3) $\frac{42}{3}$°C (4) $\frac{56}{3}$°C

Answer (4)

Sol. $\Delta T = T_2 - T_1 = 16°C$

And $T_0 = 16°C$

$$\frac{\Delta T}{t_1} = -k(32-16°) \quad ...(i)$$

$$\frac{(24-T_3)}{4} = -k\left(\frac{24+T_3}{2} - 16\right) \quad ...(ii)$$

$$\frac{16}{4} = -k(16)$$

$$\Rightarrow \frac{(24-T_3)}{4} = -k\left(12+\frac{T_3}{2}-16\right)$$

$$\Rightarrow \frac{16}{24-T_3} = \frac{16}{\frac{T_3}{T_2}-4}$$

$$\Rightarrow \frac{T_3}{2} - 4 = 24 - T_3$$

$$\Rightarrow \frac{3T_3}{2} = 28$$

$$\Rightarrow T_3 = \frac{56}{3}°C$$

27. A solid sphere is rolling without slipping on a horizontal plane. The ratio of the linear kinetic energy of the centre of mass of the sphere and rotational kinetic energy is :

(1) $\frac{5}{2}$ (2) $\frac{4}{3}$

(3) $\frac{3}{4}$ (4) $\frac{2}{5}$

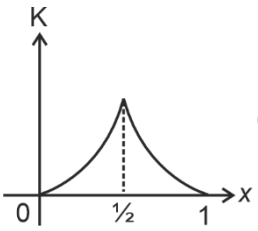
Answer (1)

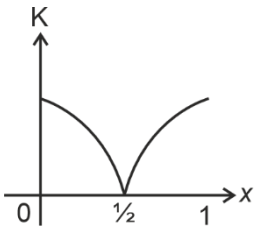
Sol. $KE_{(T)} = \frac{1}{2}mv^2$

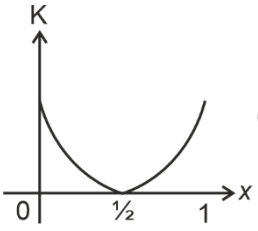
$$KE_{(R)} = \frac{1}{2}\cdot\frac{2}{5}mR^2\cdot\frac{v^2}{R^2} = \frac{1}{2}mv^2\left(\frac{2}{5}\right)$$

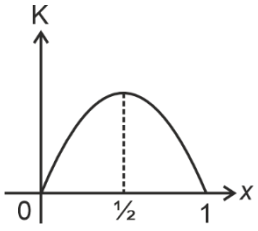
So, $\frac{KE_{(T)}}{KE_{(R)}} = \frac{5}{2}$

28. A particle oscillates along the *x*-axis according to the law, $x(t) = x_0 \sin^2\left(\frac{t}{2}\right)$ where x_0 = 1 m. The kinetic energy (K) of the particle as a function of *x* is correctly represented by the graph

(1)


(2)


(3)


(4)


Answer (4)

